

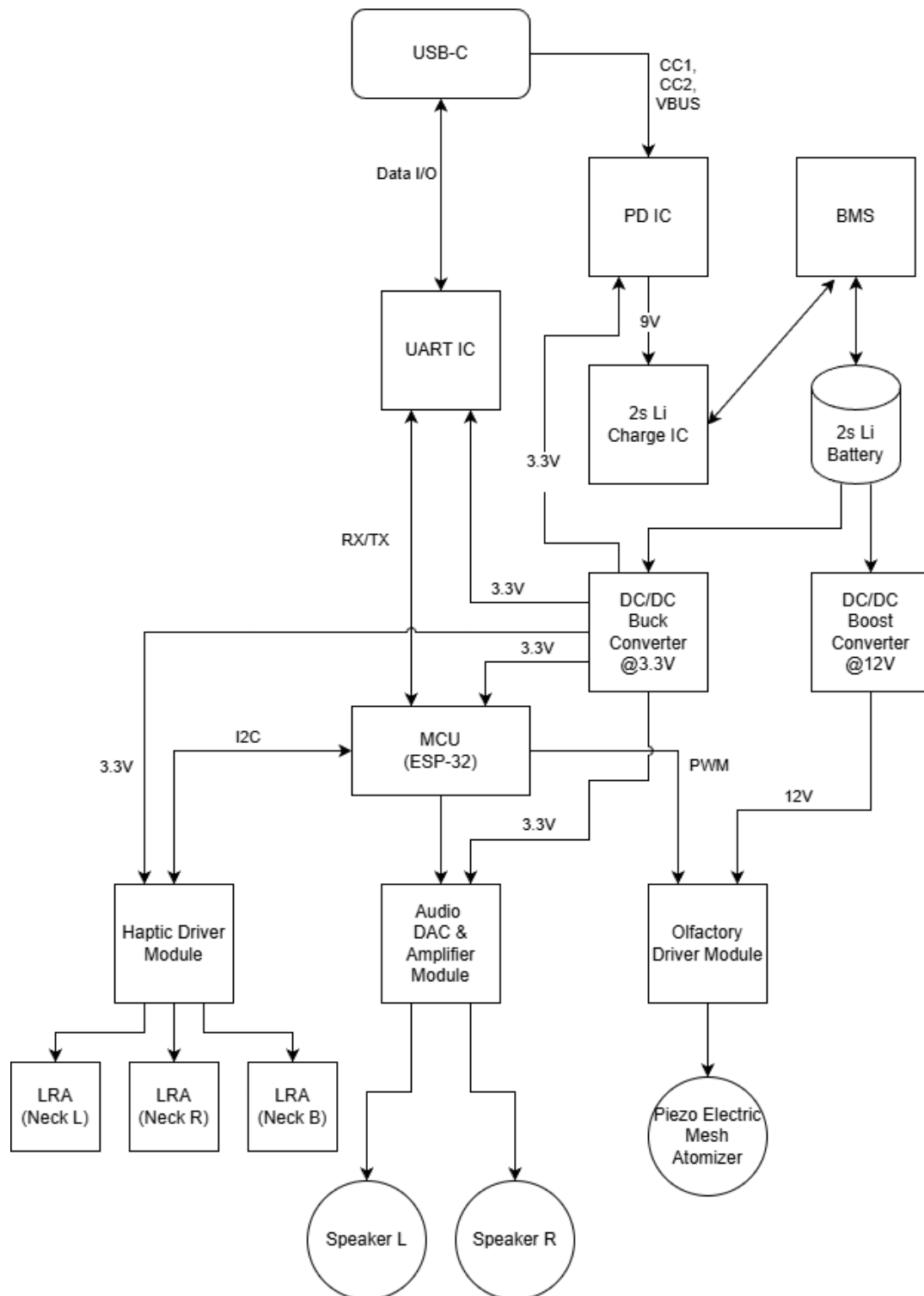
AI Deputy – Electronics Design

Brainstorming & Notation Document

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(OLD) System Overview



Charge IC never got implemented due to a communication mismatch regarding task allocation between sub-teams, I thought this was being handle by them.

GitHub Repositories

The software is developed using the **PlatformIO** environment in **Visual Studio Code**. You can either copy the *main.c* file into your preferred development environment or install **PlatformIO** extension in **Visual Studio Code** to build and run the project directly.

Included in the repo is the Schematic and PCB design. Along with footprint choices for individual components. The design software is KICAD.

GitHub Repository: https://github.com/zayinconde/Expert_In_Teams.git

Design Requirements

Note: These are the relevant design requirements for electronics provided by the stakeholders.

a. Sound (speakers) on both sides

- Speakers on both sides are not a problem and will be done.

b. suitable receiver with low latency to communicate with the AI-Deputy web app (most likely Bluetooth)

- NOTE: Initially the project was going to use Bluetooth BLE (low energy) which has a VERY low latency (about 6ms). Due to the design change Bluetooth classic is the only option to support audio transmission. However, Bluetooth classic has 100-150ms of latency. Additionally, another 50-100ms for LRM rise time and the scent emitter will possibly be even longer but must be tested. (enabling boost converter time + time to emit, possibly fixed in code?)

c. The connectivity design must prove that all mobile devices on the market will

connect and can use the features of the collar. Very old devices - like older than ~ 7 years – may be excluded from that rule.

- Bluetooth 😊

d. For the haptics a vibration actuator is placed in both sides. The design must ensure that the vibration function is transmitted gentle but predictable to the skin of the person who wears the collar. That requirement must include variations of neck shape of different users.

e. Only one Bluetooth emitter for all three functions: sound, haptics, scent.

- Noted.

i. For the scent experience one scent would be sufficient. There is however a constraint we have not discussed so far and Jörg pointed this out today.

We need to make sure that **the scent which is emitted will disappear after the emission is stopped in a predictable time. That's certainly something which needs to be tested, including an appropriate scent.** It might even involve an active neutralization.

I can imagine it could make sense to test that short term with the e-scent lab prototype which Ventsislav showed during the first Friday meeting. You can get in touch directly with him. Also involve Jörg in the process. He can explain more details.

- Yes, we will need extensive testing. Piezo transducers datasheets lack data and are not specifically designed for the purpose of emitting scent. So no data is publicly available on this. Many factors impact this outcome.

j. The scent emitter can be right or left. Choose whatever is best for the internal design. If it's convenient to build the whole device internally fully symmetric it's Ok if the device has the potential (optional for later) to add a second emitter on the opposite side.

- Noted.

k. The battery does not need to hold for more than 2 hours.

- Noted.

l. One or more batteries (in case different batteries for the three features involved are of advantage).

- Doesn't make sense, adding more batteries and control circuits for said batteries will take more space than buck/boost converters.

m. Sound volume regulation (should be optional because it should be possible to use the phone. But please double check).

n. USB charging port. USB C might be the most suitable. Please check.

- Due to EU common charger directive, USB-C is the *only* option if you plan to create a product for the EU market.

MCU choice

Microcontroller choice is somewhat based on previous research, but changing to ESP32-WROOM-32E or ESP32-WROOM-32UE for dual-mode Bluetooth. According to my research, I can use Bluetooth Classic for audio streaming over Bluetooth and BLE mode for triggering GPIO pins on MCU.

ESP32-WROOM-32E – this variant has the antenna built into the chip. However, it seems that this is not ideal for a module inside of a case.

ESP32-WROOM-32UE - without antenna, but has built-in impedance matching and uses u.FL antenna connection.

Takeaway – I will need to find (50ohm?) dual dual-band antenna for the module. Also, I don't forget to find an external flash chip if necessary. It appears there is already 4MB of flash on the chip.

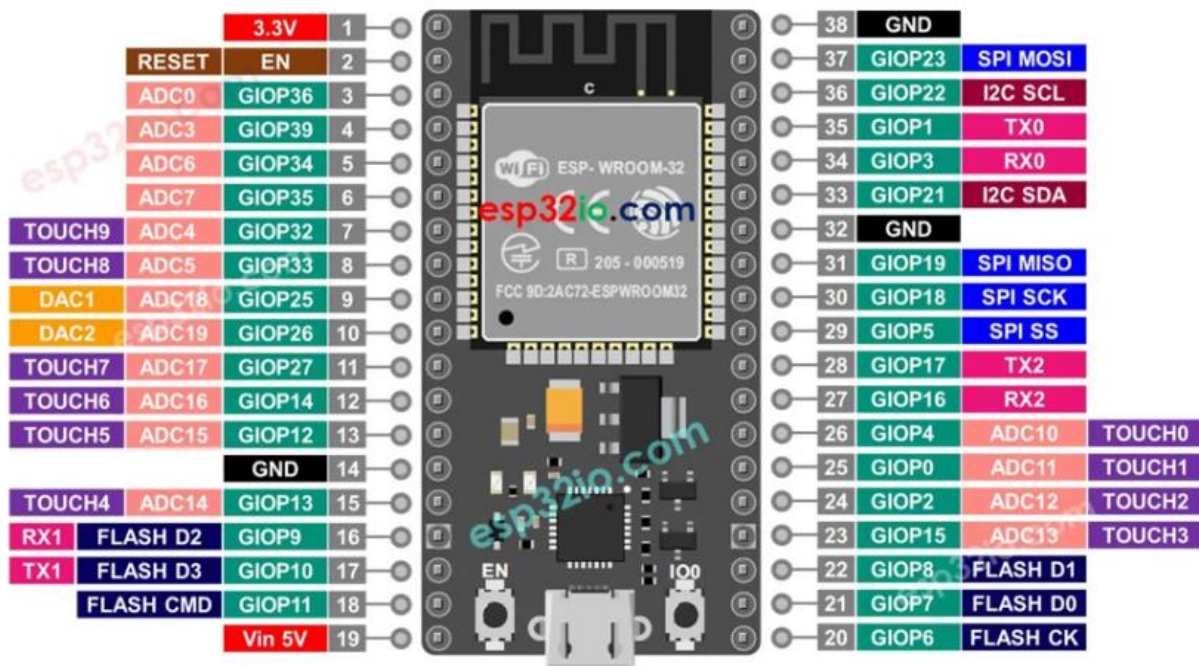
The sleep current of the ESP32 chip is less than 5 μ A, making it suitable for battery-powered and wearable electronics applications.

Datasheet: <https://asset.conrad.com/media10/add/160267/c1/-/en/002383844DS00/datablad-2383844-espressif-esp32-wroom-32ue-m113eh3200uh3q0-tradlost-modul-1-stk.pdf>

Product link: <https://www.conradelektronik.dk/da/p/espressif-esp32-wroom-32ue-m113eh3200uh3q0-traadlost-modul-1-stk-2383844.html>

Dev. Board Link: <https://dk.rs-online.com/web/p/mikrokontroller-udvikling/2863991?gb=s>

- https://www.reddit.com/r/diyelectronics/comments/112dx6n/esp32_38_pin_pinout_cheat_sheet/#lightbox
- https://web.archive.org/web/20240208091140/https://docs.espressif.com/projects/esp-idf/en/latest/esp32/_images/esp32-devkitC-v4-pinout.png



Module	Chip embedded	Flash	Module dimensions (mm)
ESP32-WROOM-32E (PCB)	ESP32-D0WD-V3	4 MB ¹	18.0 × 25.5 × 3.1
ESP32-WROOM-32UE (IPEX)			18.0 × 19.2 × 3.2

Notes:

1. The module with 8 MB flash or 16 MB flash is available for custom order.
2. For detailed ordering information, please see [Espressif Product Ordering Information](#).
3. For dimensions of the IPEX connector, please see Chapter 7.3.

6 Peripheral Schematics

This is the typical application circuit of the module connected with peripheral components (for example, power supply, antenna, reset button, JTAG interface, and UART interface).

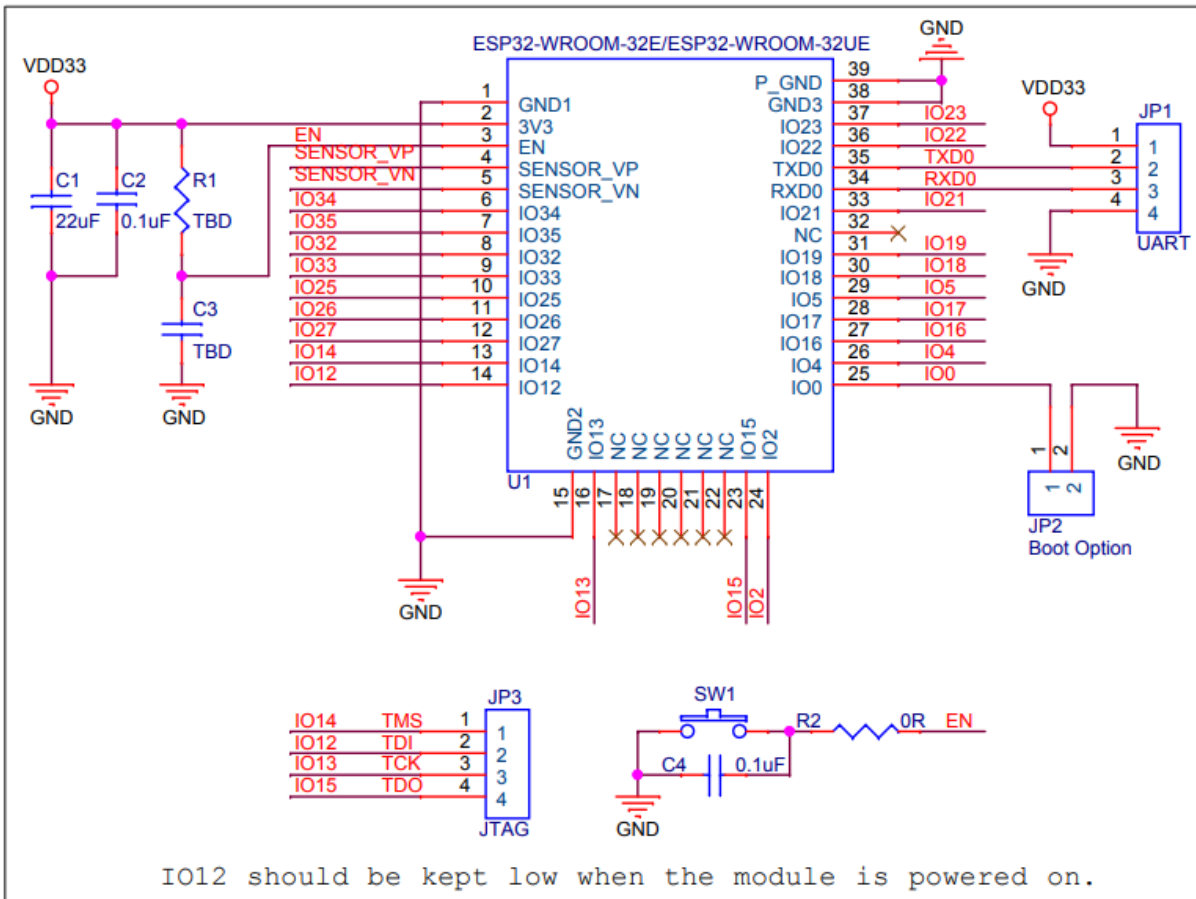
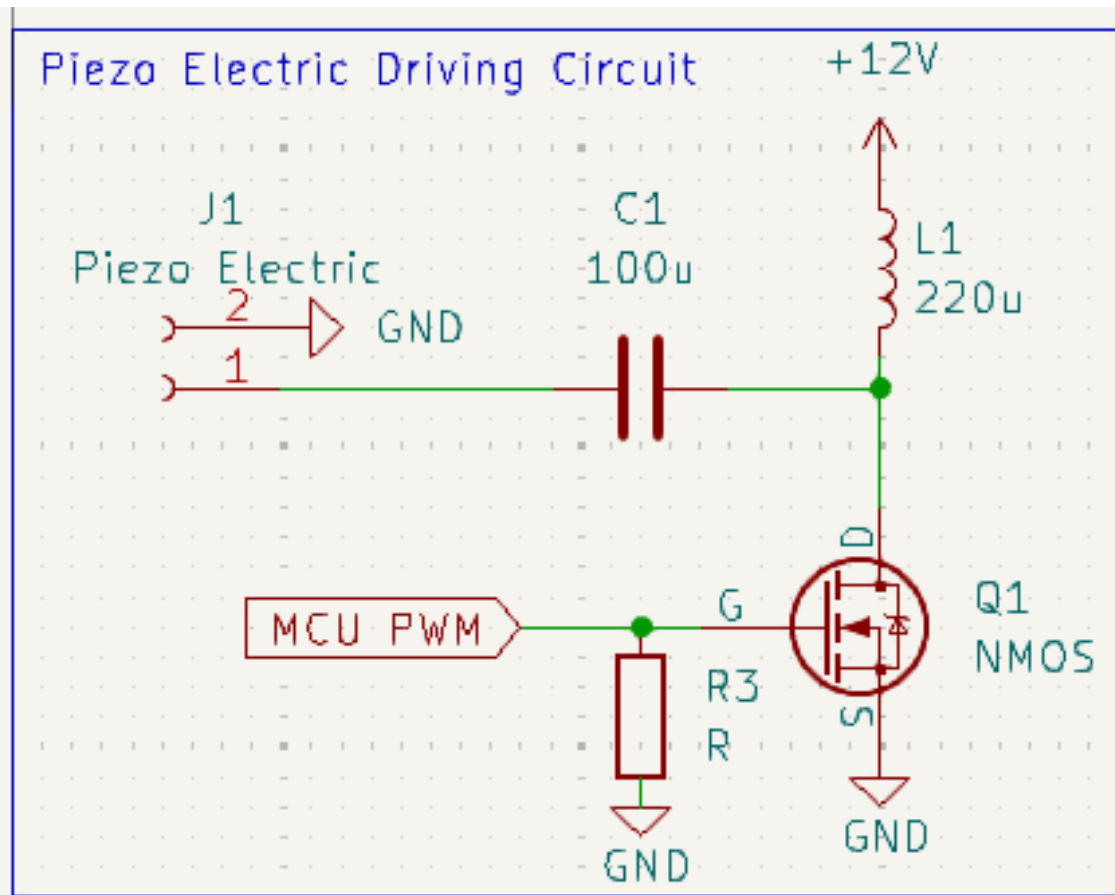


Figure 6: ESP32-WROOM-32E & ESP32-WROOM-32UE Peripheral Schematics

Olfactory system



Component List

- Piezo electric mesh transducer (ordered 25mm)
 - We are working on getting 12mm (due to neck design change)
- 100μF Capacitors (footprint unknown atm)
 - For lab testing I will find something in the lab
 - Will order proper SMD components once its pasted testing stage
- 220μH inductor
 - For lab testing I will find something in the lab
 - Will order proper SMD components once its pasted testing stage

- A 3.3 V PWM logic signal drives the gate of an NMOS transistor, which acts as a low-side switch to pulse a 12 V supply to the load. (level shifting).
 - For testing I will use BS170 from lab. Will find a more adequate NMOS later. (lower $R_{ds(on)}$, true 3.3V logic gate drive)
 - Note: The piezo 25mm res. frequency is 1.7MHz, I am concerned a simple low side transistor is not going to be sufficient and the waveform may collapse way before in the kHz range. I might have to contact Ventsislav Ivanov in regards to how they got success with a low side switch.
 - IRLZ44NPBF has lower $r_{ds(on)}$ and true logic level gate.
 - <https://dk.rs-online.com/web/p/mosfet/5410086?gb=s>
 - <https://docs.rs-online.com/de24/0900766b807913d7.pdf>
 - https://dk.farnell.com/infineon/irlml6344trpbf/mosfet-n-ch-30v-5a-sot23/dp/1857299?gross_price=true&CMP=KNC-GDK-GEN-Shopping-Pmax-High-ROAS-Test-1333&mckv=_dc|pcrid||&gad_source=1&gad_campaign_id=20746710150&gbraid=0AAAAAD8yeHkZ9UJ8ixbJJ_gMQwETeaWFG&gclid=CjwKCAjwx fjGBhAUEiwAKWPwDjCSMFpyDK1neoN-_9jc9NimH1ICmGXSM7zTVAbroG1YuAzzigZ26hoCoO4QAvD_BwE
 - SMD variant above for PCB

Olfactory Simulations

syms T

$f = 1/T$

$f = 1/(588 \cdot 10^{-9})$ % almost 1.7Mhz (potential resonant frequency for piezo electric transducer)

$f =$

$\frac{1}{T}$

$f = 1.7007e+06$

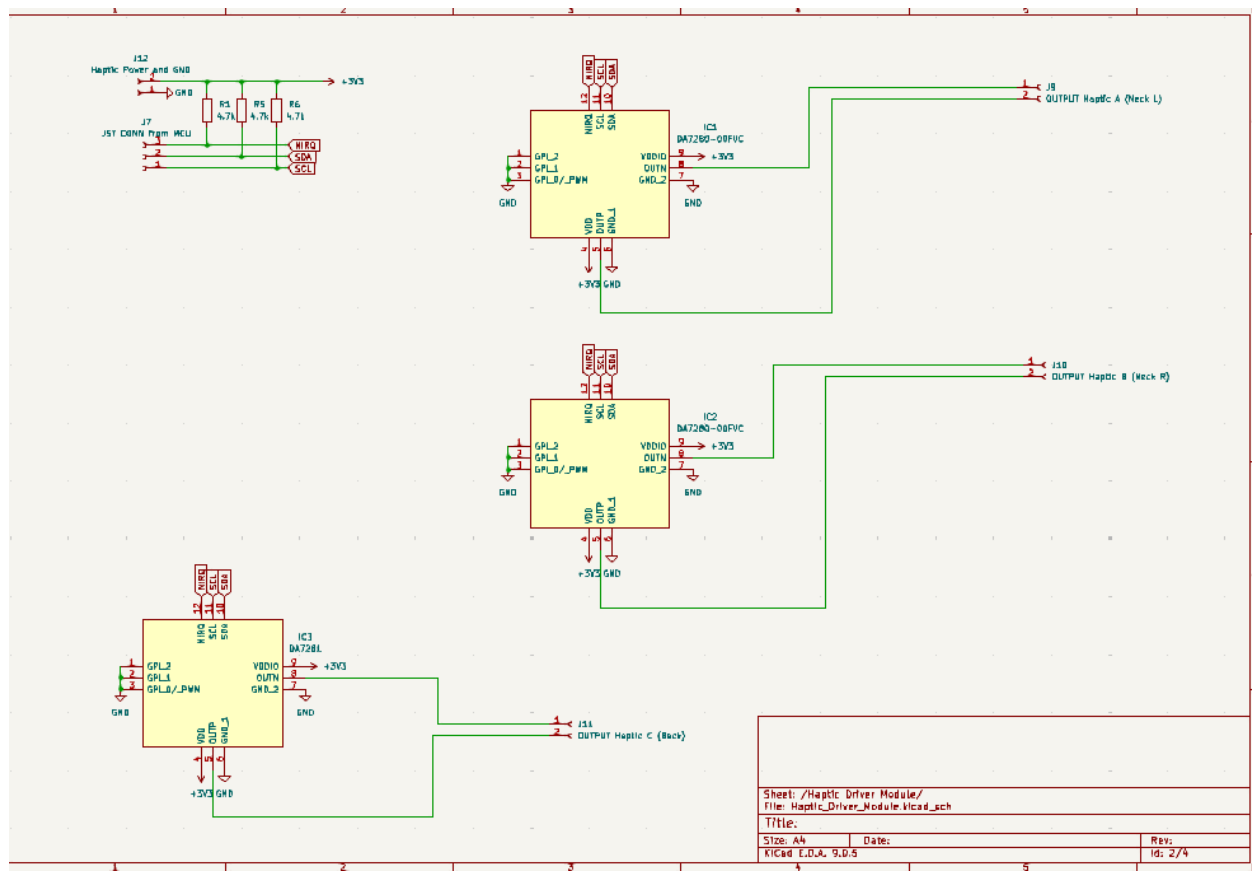
Found a piezo electric atomizer from amazon that is 16mm and has a much lower resonant frequency (110kHz) this is used along with Ventis design.

Haptic System

Small Solution Footprint

Available in an ultra-small 1.35 mm x 1.75 mm, 0.4 mm pitch, 0.545 mm height, 3 x 4 WLCSP, or a 3.0 mm x 3.0 mm, 0.65 mm pitch, 0.78 mm height, 12 lead QFN package, DA7280 minimizes the required PCB size and overall solution cost. In the typical application case, only a single 100 nF decoupling capacitor is required. See [Section 9](#) and [Section 10](#).





IMPORTANT NOTE:

I need to order DA7280, DA7281, DA7282 each version is the same but it has a different I2C address burned into the chip. Which I need because I am using 3 of them.

Audio Playback Circuit

IS31AP4991A-GRLS2-TR (changed mind, not using)

<https://www.digikey.dk/en/products/detail/lumissil-microsystems/IS31AP4991A-GRLS2-TR/5319733>

- Class AB (inefficient compared to class d)
- 2-channel stereo driver
- 1.1W @ 5V, [0.6-0.8@3.3W](#)
- Mono (need 2 preferably 4 if issues arise)
- Supports 8 ohm impedance load

- Simple circuit application (single ended)

Datasheet: https://www.lumissil.com/assets/pdf/core/IS31AP4991A_DS.pdf

IS31AP2005-SLS2-TR

<https://www.digikey.dk/en/products/detail/lumissil-microsystems/IS31AP2005-SLS2-TR/2863804>

- Class d
- Mono (need 2 preferably 4 if issues arise)
- Supports both 4 and 8 ohm loads
- 1.5W at 4 ohm 3.3v
- 0.8w at 8 ohm 3.3V
- 35k stock

Datasheet: <https://www.digikey.dk/en/products/detail/lumissil-microsystems/IS31AP2005-SLS2-TR/2863804>

PAM8302AASC4

<https://www.digikey.dk/en/products/detail/diodes-incorporated/PAM8302AASCR/4033281>

- Class d
- 4 and 8 ohm loads
- 3.3V around 1W 4 ohm
- 3.3V around 0.6w ohm
- 69k stock

Speaker Options:

- <https://dk.farnell.com/en-DK/pui-audio/as01508ms-sc11-wp-r/speaker-600hz-20khz-8ohm-89db/dp/4411340>
 - Very small but can only reach 600hz before significant fall off. (most adult male voice spectrum 80Hz -20kHz) (res. F 850Hz)
 - Very small (Dimensions: 16mm x 9mm x 3mm)
 - IP67 rating dustproof and waterproof. Designed for mobile application.
 - 91dB output
 - Poorer sound quality but better battery lifespan.
 - Z=6 ohm
 - Datasheet: <https://www.farnell.com/datasheets/4197292.pdf>

- <https://dk.farnell.com/en-DK/pui-audio/as04004po-2-lw152-wr-r/speaker-200hz-20khz-4ohm-82dba/dp/4411371>
 - Larger (dimensions: 40mm x 28.5mm x 11.7mm)
 - Lower output but if placed next to ear will be sufficient - 82dB
 - Better frequency response 200Hz-20kHz (res. f 420Hz)
 - Not as good water and dust resistance (no IP rating)
 - Input 3W, Max 4W
Needs 3-5W amplifier.
 - Better sound quality at the cost of battery lifespan.
 - Z=4 ohm
 - Datasheet: <https://www.farnell.com/datasheets/4197206.pdf>
 - <https://dk.farnell.com/diodes-inc/pam8304asr/audio-power-amp-d-40-to-85deg/dp/3373822>
 - <https://dk.farnell.com/en-DK/diodes-inc/pam8404kgr/audio-power-amp-d-40-to-85deg/dp/3483196>
 - <https://dk.farnell.com/en-DK/diodes-inc/pam8407dr/audio-power-amp-d-40-to-85deg/dp/3373827>

[ESP-DSP Series 2 | Building Efficient Digital Filters with FIR | SaludPCB](#)

Link above provides examples of using Espressifs DSP library for filtering (FIR filters)

System Schematic and PCB Design

Component Size:

Imperial				Size	Metric			
Resistor Case Code	Approx. Length (in)	Approx. Width (in)	Power (W)		Resistor Case Code	Approx. Length (mm)	Approx. Width (mm)	Power (W)
01005	0.016	0.008	0.031	▪	0402	0.4	0.2	0.031
0201	0.02	0.01	1 / 20 (0.05)	▪	0603	0.6	0.3	1 / 20 (0.05)
0402	0.04	0.02	1 / 16 (0.062)	▪	1005	1.0	0.5	1 / 16 (0.062)
0603	0.06	0.03	1 / 10 (0.10)	■	1608	1.6	0.8	1 / 10 (0.10)
0805	0.08	0.05	1 / 8 (0.125)	■	2012	2.0	1.25	1 / 8 (0.125)
1206	0.125	0.06	1 / 4 (0.25)	■	3216	3.2	1.6	1 / 4 (0.25)
1210	0.125	0.10	1 / 2 (0.5)	■	3225	3.2	2.5	1 / 2 (0.5)
1812	0.18	0.125	3 / 4 (0.75)	■	4532	4.5	3.2	3 / 4 (0.75)
2010	0.20	0.10	3 / 4 (0.75)	■	5025	5.0	2.5	3 / 4 (0.75)
2512	0.25	0.125	1	■	6332	6.3	3.2	1

Its not recommended to go smaller than 0805 for manual soldering. 0603 is possible with expert tools and experience. Smaller than 0603 needs a microscope. For this Project to mitigate risks I will not use smaller than 0805 SMD components. (not in regard to ICs)

Note: 0805 and 2012 have the same physical size component just different name due to metric and imperial systems.

Component List

Hyper Link: Bill Of Materials - BOM list.xlsx

Add power button and Bluetooth pairing button!

Power Delivery Controller design

CYPD3177 PD trigger was chosen for the design due to its small footprint and often used in mobile devices.

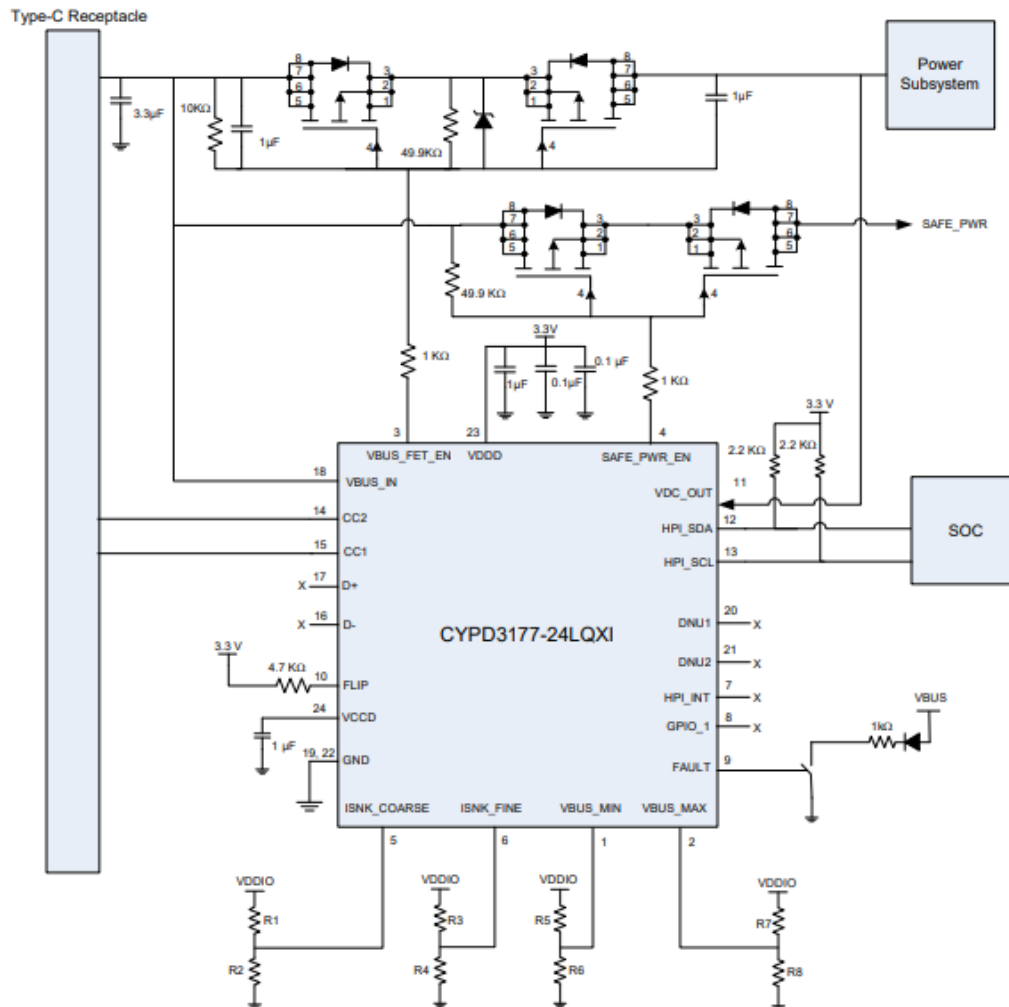
CYPD3177 typical application:

Application Overview

Figure 3 shows the EZ-PD BCR-based application diagram using the 24-pin QFN part. It has three main parts: USB Type-C receptacle to provide the input power to the application, the Power Subsystem used as the output power, and four sets of resistor divider networks to select the desired output voltage and current values.

The 'Fault' pin is used to indicate any voltage faults. When a fault condition is enabled, the output voltage of this application will go down to 0V and the EZ-PD BCR device will attempt a protocol reset to recover from fault. For a detailed reference schematic, refer to the [CY4533 EZ-PD BCR EVK schematic](#).

Figure 3. EZ-PD BCR based Application Diagram (for Electronic Systems Requiring 12 V to 15 V Input at 2 A)



Notes

- NOTES**
1. Refer to Table 2, Table 3, and Table 4 for values of these resistor divider networks.
 2. FLIP pin is in LOW state when TypeC Plug is upside-up, and in Hi-Z state when upside-down.
 3. Use a 50-k Ω resistor on the FLIP pin to set the UFP VDO data capability bit to 0.

Back to Back PMOS design

The back-to-back PMOS configuration is used to:

- Isolate VBUS from the system circuitry
- Prevent reverse current flow

- **Safely connect and disconnect a USB-PD triggered voltage**

This is especially important in USB-C / PD designs where:

- VBUS voltage may be **5 V, 9 V, 12 V, 15 V, or 20 V**
- The system must **not back-feed** VBUS
- Power should only be applied **after PD negotiation**

Two **P-channel MOSFETs** are placed **in series** on the high side (VBUS), with their:

- **Sources connected together**
- **Drains facing outward** (opposite directions)
- **Gates tied together**

This arrangement makes the body diodes face **opposite directions**.

Why Two PMOS Instead of One?

A single PMOS has an internal **body diode** that always conducts in one direction, even when the MOSFET is off.

Using **two PMOS back-to-back**:

- One body diode blocks current in one direction
- The other blocks current in the opposite direction
- Result: **no current flow when the switch is off**, in either direction

This gives **true bidirectional isolation**.

Operating Principle

1. Switch OFF (Isolation Mode)

- Gate voltage $\approx$ Source voltage
- Both PMOS transistors are **OFF**
- Body diodes oppose each other
- **No current can flow** from VBUS to system or from system to VBUS

This is the default safe state before PD negotiation.

2. Switch ON (Power Enabled)

- Gate is pulled **low** relative to the source
- Both PMOS transistors turn **ON**
- Channel conduction bypasses body diodes
- VBUS is connected to the system with **low resistance**

This happens only after the PD trigger confirms a valid voltage.

Typical Use in a PD Trigger Circuit

1. USB-C PD controller negotiates voltage
2. Correct VBUS level is detected (e.g. 12 V)
3. Control signal pulls PMOS gates low
4. Back-to-back PMOS connects VBUS to system load
5. If cable is removed or fault occurs → gates released → isolation restored

Advantages

- Prevents **reverse current**
- Protects PD source and system
- Works for **high-side switching**
- Simple and reliable
- No active IC required (discrete solution)

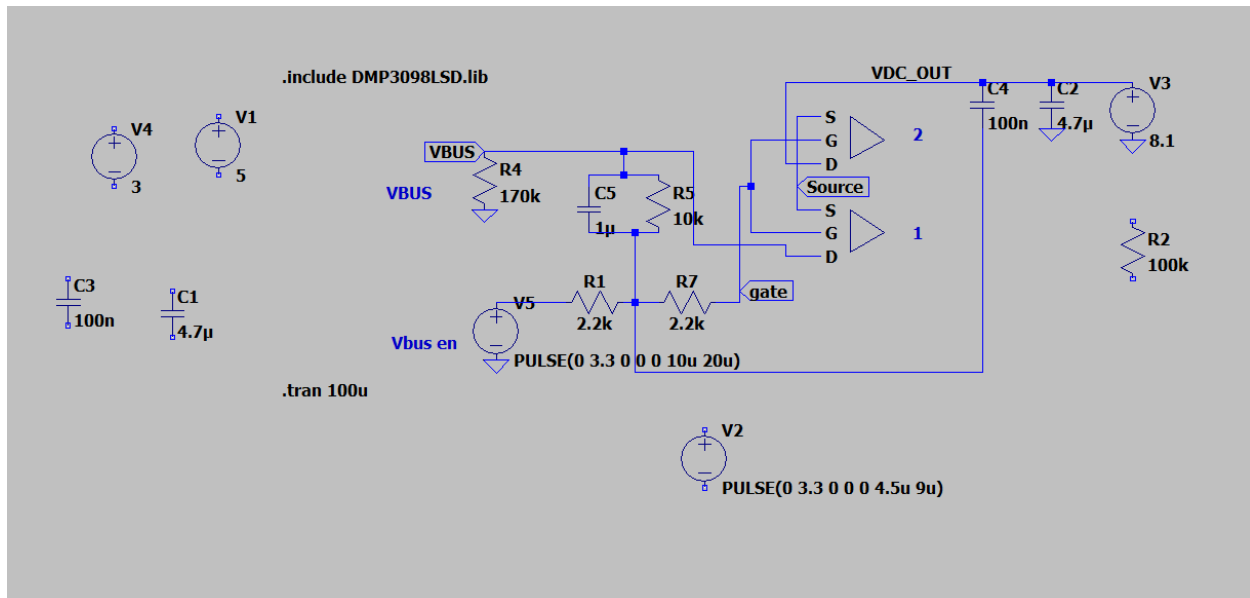
Design Notes (Practical)

- PMOS **V_{GS} rating** must handle max VBUS
- Choose **low $R_{DS(on)}$** for efficiency
- Gate resistor often added for controlled turn-on
- Gate pull-up ensures OFF by default

The back-to-back PMOS design acts as a bidirectional high-side load switch.

It allows VBUS to be safely connected only when intended, while fully isolating the system at all other times—making it ideal for USB-PD trigger applications.

Simulations were carried out in LT spice to further understand the design:



Appendix

Note: All research until this point was based off of the handheld design. The handheld design has now been changed to a “neck” type of design. Due to the significant change, the following research represents early design exploration for the previous form factor and is retained here for documentation and traceability.

Expert In Teams Project – Electronic options

Microcontroller

- ESP32
 - Various models for consideration
 - ESP32 C3 Mini
 - ESP32-C3 (BLE)
 - ESP32 WROOM 32E
 - S3
 - **PROS**
 - Build in Bluetooth
 - No RF matching or antenna tuning
 - Simpler and smaller PCB footprint
 - Good eco system with libraries
 - 3.3V
 - Cheap
 - **CONS**
 - Higher active current draw during BLE use
 - Overkill for a simple I/O system
 - Larger module size compared to RP chip
- RP2040
 - **PROS**
 - Even cheaper than esp32
 - Dual core arm, good for deterministic tasks
 - PIO blocks – great for precise /custom I/O (not relevant for this project)

- Very low idle power
- **CONS**
 - No built-in wireless Bluetooth (and Wi-Fi)
 - Needs external flash and crystal
 - Less ram and flash
 - Makes firmware stack more complex with communications between BLE and MCU
- nRF52
 - **PROS**
 - Built-in Bluetooth
 - Much lower power consumption compared to ESP32
 - Common in wearables
 - Can be powered directly from coin cells and Li-ion with tiny regulators
 - Tons of BLE profiles, good mobile integration
 - **CONS**
 - Weaker CPU performance
 - Dev environment is more complex (Nordic SDK, Zephyr, Arduino-nRF52)
 - Smaller ecosystem
 - Chips are QFN/BGA only, difficult to solder unless you buy a nRF52840

Conclusion

It appears that the ESP32 has the lowest risk of damaging when soldering, is the easiest and most well-known so lots of support can be found online. I have personally never used an esp32 so this will be a learning curve for me. Additionally, the ESP32 seems like a good microcontroller to be familiar with due to its wide range of applications and its Bluetooth and Wi-Fi capabilities.

It does appear that the nRF52 would be a better solution for a battery powered handheld due to its low power consumption in comparison to the ESP32. However, the high risk of a mistake during the soldering process and its low online support is the reason I will not be choosing this microcontroller for this project.

Power Supply

- **Considerations**

- rechargeable Li-ion /Li-po
 - According to CE certifications

Haptic Feedback

- **Eccentric Rotating Mass (ERM) motors**
 - DC motor with off center weight
 - **PROS**
 - Cheap
 - easy to implement
 - widely available
 - **CONS**
 - Only basic haptic feedback
 - Slow response
 - Like in phones and console controllers
- **Linear Resonant Actuators (LRA)**
 - A voice coil moves a small weight at a resonant frequency
 - **PROS**
 - more crisp haptic feedback
 - faster response
 - **CONS**
 - needs a driver
 - resonant frequency tuning

Haptic conclusion

A mix of ERM (eccentric rotating mass motor) and LRA (linear resonant actuator). ERM for strong feedback and LRA for crisp taps, clicks and pulses.

Note: ERM needs simple mosfet or motor driver pwm and LRA needs a proper driver (e.g., TI DRV2605L or DRV2625) to excite it at resonant frequency.

Considerations: is there an all in one solution? The motors cannot be found in an all in one solution however the drivers can be.

- **Piezo Haptic Actuators**

- **Electroactive Polymer**
- **Custom mechanical clicks**

Scent emitter

Piezo electric mesh with deionized water and mineral oils (aroma)

Driving circuitry pretty easy? Already given.

Feedback LED for Bluetooth

Feedback LED for device on

Feedback LED low battery

Addressable “mood” LEDs? (no)

40Mhz Crystal capacitance calculations.

```
% Crystal Capacitance Calculations
c1=27e-12
c2=27e-12
Cstray=5e-12 % random capacitance of pcb, traces, grounds planes etc)(most designers assume

CL=((c1*c2)/(c1+c2)+Cstray)
% should be around 18pF according to datasheet|
```

```
c1 = 2.7000e-11
c2 = 2.7000e-11
Cstray = 5.0000e-12
CL = 1.8500e-11
```

Load capacitance (CL)	-----	18.0	-----	pF	Standard (See options if other than STD)
Drive Level	-----	10	100	μW	
Aging	-3	-----	+3	ppm	@25°C±3°C First year
Insulation Resistance	500	-----	-----	MΩ	@ 100Vdc ± 15V

PCB current status

